

Gold Star: ADHD Routine Improvement Application

CS 411 21378 – Lab 1 Version 1

Gold Star: ADHD Routine Improvement Application

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1 Introduction

Individuals with attention-deficit/hyperactivity disorder (ADHD) or ADHD-like traits often experience difficulties with developing and maintaining consistent routines. ADHD is medically classified into three subtypes: Inattentive, Hyperactive-Impulsive, and Combined (Chhabildas et al., 2001, 5229). ADHD can involve productivity-altering traits such as inattention, hyperactivity, impulsivity, memory problems, and a lack of focus (Staley et al., 2026, p890; Advanced Psychiatry Associates, 2026).

ADHD often persists beyond childhood, and the difficulties associated with maintaining routines can continue into adulthood. Routines may last anywhere from three days to “part of a month” and can be vulnerable to persistent Routine Adaptations (ADHD Philadelphia, 2026). Routine Adaptations can lead to frustration, anxiety, avoidance, and reduced self-efficacy (Chen et al., 2026, pp. 2, 7). Creating new schedules also requires individuals to spend time determining tasks and estimating uncertain timeframes. Additionally, individuals may forget to review their schedules, or Task Deviations may occur.

Gold Star is a web-based habit-building and time management platform tailored to individuals affected by ADHD. The platform is designed to help users organize schedules, break down tasks, and build sustainable routines. By providing structure while allowing users to adapt their schedules, Gold Star is intended to make it easier for users to start tasks, stay on track, and follow through with their daily routines.

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2 Gold Star Product Description

Gold Star is a web-based productivity and habit-building platform designed primarily for individuals with ADHD who struggle with routine creation and maintenance. Individuals with ADHD may experience difficulties with organization, time management, and keeping up with daily responsibilities. Gold Star addresses these challenges by providing users with a structured environment for managing tasks, schedules, and habits.

Gold Star provides a low-stakes environment in which users can monitor their tasks, schedules, and habits without becoming overwhelmed. The platform is designed to support routine development and maintenance while allowing users to make change as their needs change. This approach allows users to maintain their schedules while also adapting o missed tasks and changing priorities.

The overall purpose of Gold Star is to make a routine development and maintenance more manageable for users who struggle with long-term goal consistency. The platform combines scheduling, task management, habit-building techniques, and productivity features to provide users with a system for organizing their daily activities.

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2.1 Key Product Features and Capabilities

Gold Star focuses on helping users organize tasks and schedules through an Adaptive Scheduling system. The system adjusts to missed tasks and changing priorities, allowing the schedule to change as the user's circumstances change. This flexibility is intended to reduce the frustration associated with falling behind and make it easier for users to continue working toward their goals.

Gold Star also incorporates productivity techniques such as Habit Stacking, activity Alternation, and the Pomodoro Technique. These techniques are intended to support users who struggle with ADHD-like symptoms by providing different approaches for organizing and completing activities.

Users can select their productivity level and modify their schedules throughout the day. This flexibility allows users to adjust their schedules based on their current needs. Users can also group tasks according to importance, allowing the platform to be personalized to individual scheduling needs.

The platform rewards users for completing tasks and reaching milestones. However, Gold Star does not punish users for missing goals. Instead, the platform includes a "Roll Over" system that allows users to move missed tasks to another day. This feature allows users to continue using their schedules without treating a missed task as a failure of the entire routine.

By combining flexibility and structure, Gold Star is designed to help users address common ADHD-related challenges involving task initiation, consistency, commitment, and completion. These features are intended to support productivity while reducing the daily stress associated with managing routines.

2.2 Major Components (Hardware/Software)

Gold Star does not require specialized hardware. Users can access the application through an off-the-shelf device with a modern web browser. The server side of the application runs on a single hosting environment that supports the Node.js web server and PostgreSQL database. Deployments are automatically pushed through GitHub.

As shown in Figure 1, Gold Star follows a three-tier structure consisting of Presentation Layer, and Application Layer, and a Data Layer. The Presentation Layer consists of the web-based user interface. The Application Layer consists of the Node.js web server and API, including authentication and notification handler. The Data Layer consists of the PostgreSQL database, which stores user data, tasks, and routine data.

The software to be developed includes a user interface and user experience (UI/UX) built with HTML, CSS, and JavaScript. The interfaces will include a calendar view, routine dashboard, productivity level system, and statistics dashboard. The Node.js web server will handle routing, application logic authentication components, and a WebSocket-based notification handler. The PostgreSQL database will store user account information, saved tasks, routine data, and sensitive user information in an encrypted form.

Development of Gold Star is supported by several off-the-shelf software development tools. Visual Studio Code is used as the development environment, while Git and GitHub are used for version control and code management. GitHub Actions is used for continuous integration and continuous deployment (CI/CD). Jest is used for testing, and JSDoc is used for software documentation.

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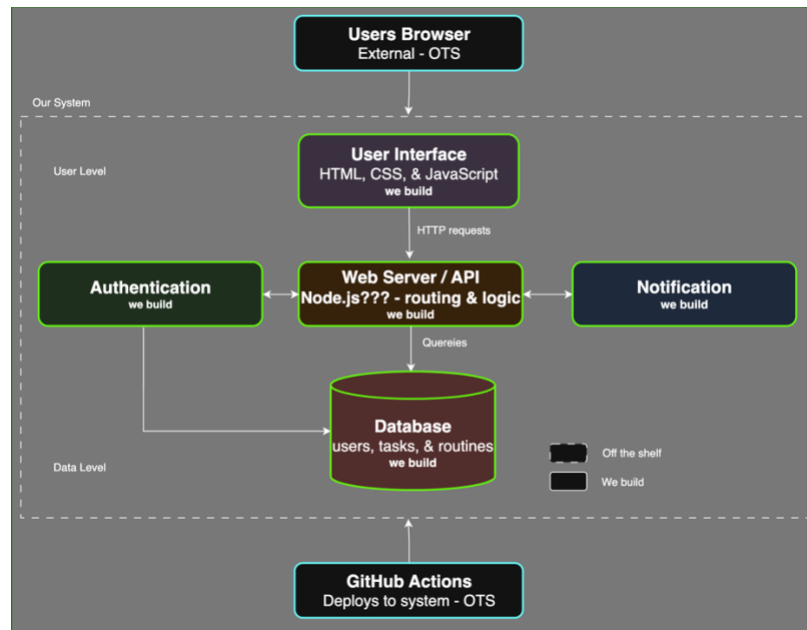


Figure 1: Gold Star's Major Functional Component Diagram.

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3 Glossary

Activity Alternation – Switching from an activity requiring high mental focus to an activity requiring low mental focus.

Activity Switching – The cognitive process of disengaging from one task and shifting mental focus to another task.

Adaptive Scheduling – To schedule a task which can be completed anytime within a set timeframe. For example, a task which can be completed anywhere inclusively between 7:00 P.M. and 10:00 P.M.

Application Programming Interface (API) - A set of standard rules, protocols, and definitions allowing different software applications to interact with one another.

Attention-Deficit/Hyperactivity Disorder (ADHD) - A common neurodevelopmental disorder that alters the brain's ability to plan, focus, remember instructions, and manage emotions.

Cascading Style Sheets (CSS) - A style sheet language used to determine the visual presentation, layout, and aesthetics of a website.

Git - A digital version control system intended to preserve a history of saved modifications, additions, and deletions to software source code.

GitHub - A cloud-based platform where developers can collaborate on software code.

Gold Star - A web-based habit-building and time management platform designed specifically for individuals with ADHD.

Habit Stacking – Interspersing new habits with existing habits to make task completion easier without feeling overwhelmed.

Hyperactivity - A state of being unusually or abnormally active.

HyperText Markup Language (HTML) - The standard markup language used to create and structure webpages. Gold Star Allen et al. 9

JavaScript (JS) - A lightweight programming language used to build dynamic interactivity on

web pages.

Jest - An open-source JavaScript testing framework.

JSDoc - An API documentation generator and syntax standard for JavaScript.

Pomodoro Technique – A time-management technique which separates work into 25-minute increments and 5-minute breaks. The standard Pomodoro process repeats this technique 4 times, then a longer break between 20 to 30 minutes is taken before the process repeats.

PostgreSQL - An open-source RDBMS that stores, manages, and retrieves data.

Reinforce - To make something stronger, more effective, or more compelling by adding support.

Relational Database Management System (RBMS) - Software used to store, manage, query, and retrieve data structured in table form.

Rigid Scheduling – To schedule a high-priority task which must either be initiated at or completed before an established time. For example, a task which must start at 6:00 P.M. or must be completed by a deadline of 6:00 P.M.

Routine Adaption – A requirement for an individual to change one's schedule or create new routines.

Routine - A sequence of actions regularly followed, or a fixed program.

Task Deviation – When a task has an unexpected change regarding completion. A task may be accomplished out of order in one's schedule, or it may not be accomplished at all.

Visual Studio Code (VSCode) - A development environment used for software engineering and web development.

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4 References

- ADHD Philadelphia. (2026, May 30). Why Adults With ADHD Struggle to Stay Consistent With Routines. ADHD Philadelphia. <https://www.adhdphiladelphia.com/blog/why-adults-with-adhd-struggle-to-stay-consistent-with-routines>
- Advanced Psychiatry Associates. (2026). Strategies for Adults Living With ADHD. Addicta. <https://advancedpsychiatryassociates.com/resources/blog/strategies-for-adults-living-with-adhd>
- CDC. (2024, May 29). ADHD Across the Lifetime. Attention-Deficit / Hyperactivity Disorder (ADHD). <https://www.cdc.gov/adhd/articles/adhd-across-the-lifetime.html>.
- Chen, J., Meng, Y., & Nie, K. (2026, March). “Not Just Me and My To-Do List”: Understanding Challenges of Task Management for Adults with ADHD and the Need for AI-Augmented Social Scaffolds. arXiv. <https://doi.org/10.48550/arXiv.2603.17258>
- Chhabildas, N. Pennington, B.F. & Willcutt, E.G. (2001, December). A Comparison of the Neuropsychological Profiles of the DSM-IV Subtypes of ADHD. *Journal of Abnormal Child Psychology*, 29, 529-540. <https://doi.org/10.1023/A:1012281226028>
- Kessler, R.C., Lane, M., Stang, P.E., & Van Brunt, D.L. (2008, April 21). The prevalence and workplace costs of adult attention deficit hyperactivity disorder in a large manufacturing firm. *Psychological Medicine*, 39(1), 137-147. <https://doi.org/10.1017/S0033291708003309>
- Kreider, C. M., Medina, S., & Slamka, M. R. (2019). Strategies for Coping with Time-Related and Productivity Challenges of Young People with Learning Disabilities and Attention-Deficit/Hyperactivity Disorder. *Children (Basel, Switzerland)*, 6(2), 28. <https://doi.org/10.3390/children6020028>
- Gold Star Allen et al. 11
- Ogrodnik, M., Karsan, S., Malamis, B., Kwan, M., Fenesi, B., & Heisz, J. J. (2023). Exploring Barriers and Facilitators to Physical Activity in Adults with ADHD: A Qualitative Investigation. *Journal of developmental and physical disabilities*, 1–21. Advance online

publication. <https://doi.org/10.1007/s10882-023-09908-6>

Staley, B.S., Robinson, L.R., Claussen, A.H., Katz, S.M., Danielson, M.L., Summers, A.D., Farr, S.L., Blumberg, S.J., & Tinker, S.C. (2024). Attention-Deficit/Hyperactivity Disorder Diagnosis, Treatment, and Telehealth Use in Adults — National Center for Health Statistics Rapid Surveys System, United States, October–November 2023. U.S. Centers for Disease Control and Prevention Morbidity and Mortality Weekly Report, 70(40), 890-895. <https://www.cdc.gov/mmwr/volumes/73/wr/pdfs/mm7340a1-H.pdf>

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